

Student's name _____

Student's number _____

Teacher's name _____



PLC PRESBYTERIAN
LADIES' COLLEGE
SYDNEY
1888

2008
TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks – 120

- Attempt questions 1-10
- All questions are of equal value

1	2	3	4	5	6	7	8	9	10	Total	Total
										/120	%

Question 1 (12 marks) Start a new sheet of writing paper

Marks

(a) Calculate $\sqrt{\frac{135.2}{12.3 \times 2.1}}$ correct to 2 decimal places. 2

(b) Factorise completely $5x^3 + 135$. 2

(c) Find integers a and b such that $\frac{3}{2\sqrt{2}+3} = a - \sqrt{b}$. 2

(d) Solve $|x-3| \geq 5$. 2

(e) Solve $x(2x+7) = 4$. 2

(f) Prove that 2

$$\frac{1}{\sec x - 1} - \frac{1}{\sec x + 1} = 2 \cot^2 x.$$

End of Question 1

Question 2 (12 marks) Start a new sheet of writing paper

Marks

(a) Differentiate with respect to x

(i) $x^2 \tan x$

2

(ii) $\frac{\log_e x}{x}$

2

(b) (i) Find $\int \sin 4x \, dx$

2

(ii) Evaluate $\int_0^1 \frac{x^2}{x^3+1} \, dx$

3

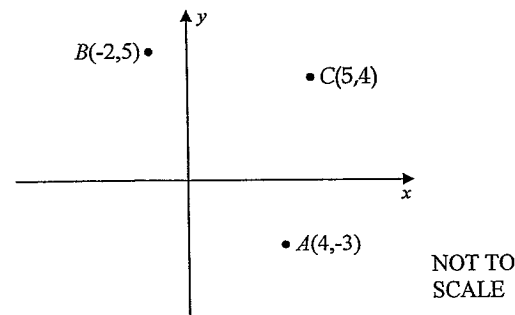
(c) Using Simpson's rule with five function values, find an approximation for $\int_0^\pi \cos^2 x \, dx$.

3

End of Question 2

Question 3 (12 marks) Start a new sheet of writing paper

Marks



In the diagram, A , B , and C are the points $(4, -3)$, $(-2, 5)$ and $(5, 4)$ respectively.

Copy or trace this diagram onto your writing paper.

- (a) Show that triangle ABC is isosceles. 2
- (b) Use the cosine rule to find the size of $\angle ACB$. 2
- (c) Find the midpoint of AB . 1
- (d) Show that the perpendicular bisector of AB has equation $3x - 4y + 1 = 0$. 2
- (e) Show that $C(5, 4)$ lies on the perpendicular bisector. 1
- (f) A circle is drawn through A and B with C as the centre.
 - (i) Find the equation of this circle. 1
 - (ii) Find the length of the arc AB . 1
 - (iii) Find the area of the segment cut off by the chord AB . 2

End of Question 3

Question 4 (12 marks)

Start a new sheet of writing paper

Marks

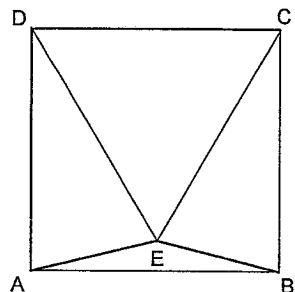
(a) Solve $3 \tan^2 \theta = 1$ for $0 \leq \theta \leq 2\pi$

3

(b) Solve $e^{2x} - 4e^x + 3 = 0$.

3

(c)



In the diagram, $ABCD$ is a square and CDE is an equilateral triangle.

(i) Copy the diagram onto your answer sheet.

(ii) Prove that $\triangle AED \equiv \triangle BEC$.

3

(iii) Hence, or otherwise, prove that $AE = BE$.

1

(iv) Find the size of $\angle AEB$. Give reasons for your answer.

2

End of Question 4

Question 5 (12 marks)

Start a new sheet of writing paper

Marks

(a) A function is defined as $f(x) = 12x - x^3$.

(i) Find the co-ordinates of the stationary point(s) of $y = f(x)$ and determine their nature.

3

(ii) Find the co-ordinates of any point(s) of inflection of $y = f(x)$

2

(iii) Sketch the graph of $y = f(x)$, clearly showing intercept(s), stationary point(s) and point(s) of inflection.

2

(iv) Find the value(s) of x for which the function, $y = f(x)$, is both increasing **and** concave up.

1

(b) Let α and β be the roots of $2x^2 - 15x + 6 = 0$.
Find the values of

(i) $\alpha + \beta$

1

(ii) $\alpha\beta$

1

(iii) $(2 - \alpha)(2 - \beta)$

2

End of Question 5

Question 6 (12 marks) Start a new sheet of writing paper **Marks**

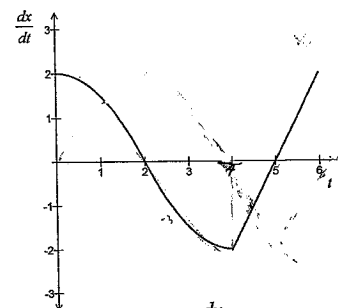
- (a) Hagrid predicts that the number of spiders in the Forbidden Forest, after t days, is given by $S = 45e^{0.26t}$.
- (i) How many spiders were initially in the forest? **1**
- (ii) At what rate is the spider population increasing after 10 days? Answer to the nearest spider. **2**
- (iii) Centaurs also live in the Forest. The number of centaurs is given by $C = 18e^{0.53t}$. How long will it take for the number of spiders to be the same as the number of centaurs? Give your answer in days, correct to 1 decimal place. **2**
- (b) Solve $2\ln(x+3) = \ln(x+7) + \ln(x+1)$. **2**
- (c) Find the values of k for which $x^2 - (k+4)x + 1 = 0$ has no real roots. **2**
- (d) The area bounded by the curve $y = \sec 2x$ between $x = \frac{\pi}{8}$ and $x = \frac{\pi}{6}$ is rotated around the x -axis. Find the volume of the solid of revolution formed. **3**

End of Question 6

Question 7 (12 marks) Start a new sheet of writing paper **Marks**

- (a) A parabola has focus $(5, -1)$ and directrix $y = 3$
- (i) Find the co-ordinates of the vertex of the parabola **1**
- (ii) Find the equation of the parabola. **2**
- (iii) The equation of the tangent to the parabola at $(-3, -7)$ has equation $2x - y - 1 = 0$. Find the exact perpendicular distance from the focus to this tangent. **2**

(b)



The graph shows the velocity, $\frac{dx}{dt}$, of a particle as a function of time, in seconds. Initially the particle is at the origin.

- (i) At what time(s) is the particle at rest? **2**
- (ii) At what time is the particle furthest from the origin? **1**
- (iii) The particle travels a distance of $2\frac{2}{3}$ cm in the first 2 seconds. Find the total distance travelled by the particle during the first 6 seconds and its final position. **2**
- (iv) Draw a sketch of the acceleration, $\frac{d^2x}{dt^2}$, as a function of time for $0 \leq t < 4$. **2**

End of Question 7

Question 8 (12 marks) Start a new sheet of writing paper

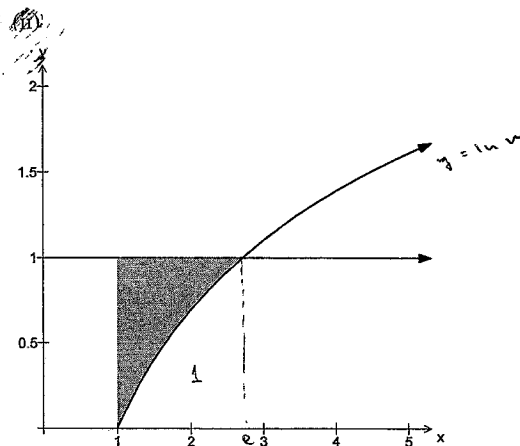
Marks

- (a) Find the equation of the tangent to the curve $y = e^{3x}$ at $x = 0$.

3

- (b) (i) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$

2



3

The shaded region in the diagram is bounded by the the curve $y = \ln x$ and the lines $y = 1$ and $x = 1$.
Using the result of part (i), or otherwise, find the area of the shaded region.

- (c) For Maths Week, Mr Palmer froze Mr Potato Head in a cylinder of ice. It melted in such a way that the block of ice remains a cylinder, similar to the original cylinder. Initially, the block of ice has a radius of 15cm and a height of 30cm. After 4 hours the ice has a volume of $5250\pi \text{ cm}^3$. The rate of change of volume, in cm^3/h , is given by

$$\frac{dV}{dt} = -k, \text{ for some constant } k > 0.$$

- (i) Show that $V = \pi(6750 - 375t)$.

3

- (ii) Find how long it took for the ice to reduce to half its original volume.

1

End of Question 8

Question 9 (12 marks) Start a new sheet of writing paper

Marks

- (a) A new PLC Year 12 memento is to be made in the shape of a sector of a circle of radius, r , and containing an angle of θ radians at the centre. The perimeter of the memento is to be 60mm.

- (i) Show that the area $A \text{ mm}^2$ is given by

2

$$A = \frac{1800\theta}{(\theta + 2)^2}$$

- (ii) 1 Show that $\frac{dA}{d\theta} = \frac{1800(2-\theta)}{(\theta+2)^3}$

1

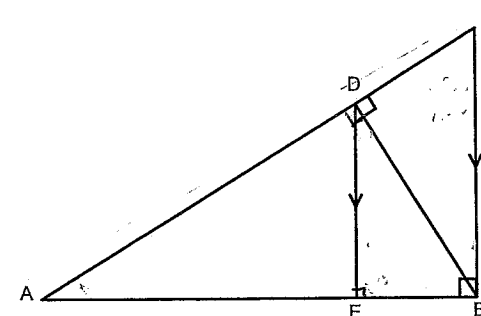
- 2 Find the maximum area of the memento.

2

- (b) Use the discriminant to show that $4x - y - 16 = 0$ is a tangent to the parabola $x^2 = 4y$.

2

(c)



$\triangle ABC$ is right-angled at B .
 BD is drawn perpendicular to the hypotenuse.
 DE is parallel to CB .
 $BC = 15\text{cm}$ and $AB = 20\text{cm}$

Find the length of

- (i) BD

2

- (ii) DE

3

End of Question 9

Question 10 (12 marks) Start a new sheet of writing paper**Marks**

A particle, A , moves along a straight line starting from 2cm to the right of the origin with an initial velocity of 2 cm/s. Its acceleration as a function of time, t_1 , is given by $a_A = -2 \sin t_1$.

A second particle, B , is initially at rest and moves along the same line starting 1cm to the right of the origin. It is initially at rest. Its acceleration as a function of time, t_2 , is given by $a_B = \cos t_2$

Particle B starts moving $\frac{\pi}{2}$ seconds after Particle A .

- (a) Show that the expressions for velocity and displacement at any time, t_1 , for particle A are given by $v_A = 2 \cos t_1$ and $x_A = 2 + 2 \sin t_1$. 2
- (b) Sketch the graph of the displacement of particle A for $0 \leq t_1 \leq 2\pi$. 2
- (c) Show that the expressions for velocity and displacement at any time, t_1 , for particle B are given by $v_B = -\cos t_1$ and $x_B = 2 - \sin t_1$. 2
- (d) Find the distance between the particles, when $t_1 = \frac{\pi}{2}$. 2
- (e) At what times do the particles meet for $0 \leq t_1 \leq 2\pi$? 2
- (f) Find the difference between the distances travelled by the 2 particles between the first 2 times the particles meet. 2

End of Paper

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Course	Mathematics 2U	Name of task/exam	Trial HSC

Question 1

(a) $\sqrt{\frac{135 \cdot 2}{12 \cdot 3 \times 2 \cdot 1}} = 2.28784 \dots$
 $= 2.29$ (to 2 dp)

(b) $5x^3 + 135 = 5(x^3 + 27)$
 $= 5(x+3)(x^2 - 3x + 9)$

(c) $\frac{3}{2\sqrt{2}+3} = \frac{3}{2\sqrt{2}+3} \times \frac{2\sqrt{2}-3}{2\sqrt{2}-3}$
 $= \frac{6\sqrt{2}-9}{8-9}$
 $= \frac{6\sqrt{2}-9}{-1}$

$= 9 - 6\sqrt{2}$
 $= 9 - \sqrt{72}$

$\therefore a = 9 \quad b = 72$

(d) $|x-3| \geq 5$

$x-3 \geq 5 \quad -(x-3) \geq 5$

$x \geq 8 \quad -x+3 \geq 5$

$-x \geq 2$

$x \leq -2$

(e) $x(2x+7) = 4$

$2x^2 + 7x - 4 = 0$

$2x^2 - x + 8x - 4 = 0$

$x(2x-1) + 4(2x-1) = 0$

$(2x-1)(x+4) = 0$

$x = \frac{1}{2} \quad x = -4$

(f) $\frac{1}{\sec x - 1} - \frac{1}{\sec x + 1} = 2 \cot^2 x$

LHS = $\frac{1}{\sec x - 1} - \frac{1}{\sec x + 1}$

$= \frac{\sec x + 1 - (\sec x - 1)}{(\sec x - 1)(\sec x + 1)}$

$= \frac{2}{\sec^2 x - 1}$

$= \frac{2}{\tan^2 x} \quad \sin^2 \theta + \cos^2 \theta = 1$
 $= 2 \cot^2 x \quad \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \cos^2 \theta} = \frac{1}{\cos^2 \theta}$
 $= \text{RHS as req'd} \quad \tan^2 \theta + 1 = \sec^2 \theta$
 $\sec^2 \theta = \tan^2 \theta$

Question 2

(a) (i) $\frac{d}{dx}(x^2 \tan x)$

$= x^2 \sec^2 x + 2x \tan x$

$= x(x \sec^2 x + 2 \tan x)$

(ii) $\frac{d}{dx} \left(\frac{\log_e x}{x} \right)$

$= \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2}$

$= \frac{1 - \log_e x}{x^2}$

b) (i) $\int \sin 4x \, dx$

$= -\frac{1}{4} \cos 4x + C$

(ii) $\int \frac{x^2}{x^3+1} \, dx$

$= \frac{1}{3} \left[\ln(x^3+1) \right]$

$= \frac{1}{3} [\ln 2 - \ln 1]$

$= \frac{1}{3} \ln 2$

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(c) $y = \cos^2 x$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1

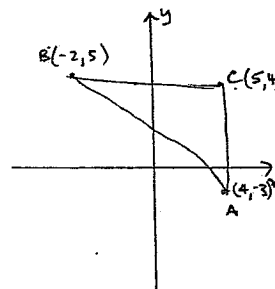
$\therefore \int_0^{\pi} \cos^2 x \, dx \quad h = \frac{\pi-0}{4} = \frac{\pi}{4}$

$= \frac{\pi}{4} \left[1 + 4\left(\frac{1}{2}\right) + 2(0) + 4\left(\frac{1}{2}\right) + 1 \right]$

$= \frac{\pi}{12} [6]$

$= \frac{\pi}{2}$

Question 3



(a) $d_{AB} = \sqrt{(4+2)^2 + (-3-5)^2}$
 $= \sqrt{6^2 + 8^2}$

$= 10$

$d_{BC} = \sqrt{(5+2)^2 + (5-4)^2}$
 $= \sqrt{7^2 + 1^2}$

$= \sqrt{50}$

$d_{AC} = \sqrt{(5-4)^2 + (4+3)^2}$

$= \sqrt{1^2 + 7^2}$

$= \sqrt{50}$

$\therefore \triangle ABC$ is isosceles, $AC = BC$

(b) $\cos \theta = \frac{50+50-100}{2 \times 50 \times 50}$

$= 0$

$\therefore \theta = \frac{\pi}{2} (90^\circ)$

(c) $\left(\frac{4-2}{2}, \frac{-3+5}{2} \right) = (1, 1)$

(d) $m_{AB} = \frac{5+3}{-2-4}$

$= \frac{8}{-6}$

$= -\frac{4}{3}$

$\therefore m_{\perp} = \frac{3}{4}$

$y-1 = \frac{3}{4}(x-1)$

$4y-4 = 3(x-1)$

$4y-4 = 3x-3$

$3x-4y+1=0$

(e) LHS = $3(5) - 4(4) + 1$

$= 15 - 16 + 1$

$= 0$

$= \text{RHS as req'd}$

(f) (i) $(x-5)^2 + (y-4)^2 = 50$

(ii) $d = r\theta$

$= \sqrt{50} \left(\frac{\pi}{2} \right)$

$= \frac{5\sqrt{2}}{2} \pi \text{ units}$

(iii) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$

$= \frac{1}{2} \cdot 50 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right)$

Solutions for exams and assessment tasks

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$$= 25 \left(\frac{\pi}{2} - 1 \right) u^2$$

Question 4.

(a) $3 \tan^2 \theta = 1$

$$\tan^2 \theta = \frac{1}{3}$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}, \frac{\pi - \pi}{6}, \frac{\pi + \pi}{6}, \frac{2\pi - \pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

(b) $e^{2x} - 4e^x + 3 = 0$

let $m = e^x$

$$m^2 - 4m + 3 = 0$$

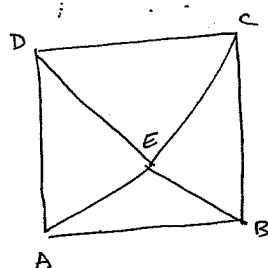
$$(m-1)(m-3) = 0$$

$$m = 1 \quad m = 3$$

i.e. $e^x = 1 \quad e^x = 3$

$$\therefore x = 0 \quad x = \ln 3$$

(c)(i)



(ii) $\angle CDA = \angle DCB = 90^\circ$ (property of square)

$$\angle CDE = \angle DCE$$
 (property of equilateral triangle)

$$\therefore \angle ADE = \angle BCE = 30^\circ$$
 (adjacent complementary angles)

in $\triangle ADE \neq \triangle BCE$

$$\angle ADE = \angle BCE (90^\circ - 60^\circ = 30^\circ)$$

$$DE = CE$$
 (equal sides of equilateral triangle)

$$AD = BC$$
 (equal sides of square)

$$\therefore \triangle ADE \equiv \triangle BCE$$
 (SAS cong test)

(ii) $AE = BE$ (corresponding sides of congruent triangles)

(iii) $AD = ED$

since $AD = DC$ &
 $ED = DC$.

(properties of squares & equilateral triangles)

$$\therefore \triangle ADE$$
 is isosceles

$$\angle ADE = \angle ADC - \angle EDC$$

$$= 90 - 60$$

$$= 30^\circ$$

$$\therefore 2\angle DAE + \angle ADE = 180^\circ$$

(angle sum of $\triangle$ is 180°)

$$\therefore 2\angle DAE = 180 - 30$$

$$= 150$$

$$\therefore \angle DAE = 75^\circ$$

$$\angle AEB + \angle DAE + \angle CED + \angle CEB$$

$$= 360^\circ$$
 (angles around a pt)

$$\therefore \angle AEB = 360 - (2 \times 75 + 60)$$

$$= 150^\circ$$

Question 5

(a) $= 12x - x^3$

(i) $= 12 - 3x^2$

$$= 3(4 - x^2)$$

for stationary pt $= 0$

$$3(4 - x^2) = 0$$

$$3(2-x)(2+x) = 0$$

$$x = \pm 2$$

When $x = -2$ $x = 2$

Test

x	-3	-2	0	2	3
$f'(x)$	-15	0	12	0	-15

$$\therefore \text{min at } (-2, -16)$$

$$\text{max at } (2, 16)$$

(ii) For pts of inflection

$$f''(x) = 0$$

i.e. $f''(x) = -6x$

$$-6x = 0$$

$$x = 0$$

test

x	-1	0	1
$f''(x)$	6	0	-6

$$\therefore \text{change of concavity}$$

so pt of inflection

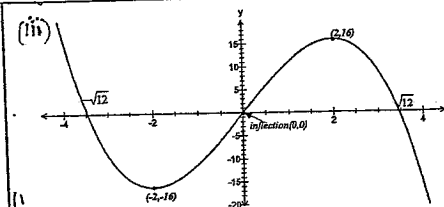
$$f(0) = 12(0) - (0)^3$$

$$= 0$$

$$\therefore \text{Pt of inflection at } (0, 0)$$

Solutions for exams and assessment tasks

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(ii) $-2 < x < 0$

b) $2x^2 - 15x + 6 = 0$

(i) $x + \beta = -(-\frac{15}{2})$
 $= \frac{15}{2}$

(ii) $\alpha\beta = \frac{6}{2}$

$$= 3$$

(iii) $(2-x)(2-\beta) = 4 - 2\alpha - 2\beta + \alpha\beta$

$$= 4 - 2(\alpha + \beta) + \alpha\beta$$

$$= 4 - 2(\frac{15}{2}) + 3$$

$$= -8$$

Question 6

(a) $S = 45e^{0.26t}$

(i) when $t = 0$

$$S = 45e^0$$

$$= 45 \text{ spiders}$$

(ii) $S = 45e^{0.26t}$

$$\frac{dS}{dt} = 45(0.26)e^{0.26t}$$

$$= 11.7e^{0.26t}$$

when $t = 10$

$$\frac{dS}{dt} = 11.7e^{2.6}$$

$$= 157.5257...$$

$$\approx 158 \text{ spiders}$$

(iii) $18e^{0.53t} = 45e^{0.26t}$

$$\frac{45}{18} = \frac{e^{0.53t}}{e^{0.26t}}$$

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$$\frac{5}{2} = e^{0.27t}$$

$$0.27t = \ln\left(\frac{5}{2}\right)$$

$$t = \frac{\ln\left(\frac{5}{2}\right)}{0.27}$$

$$= 3.393669377$$

$$\approx 3.4 \text{ days}$$

$$(b) 2 \ln(x+3) = \ln(x+7) + \ln(x+1)$$

$$(x+3)^2 = (x+7)(x+1)$$

$$x^2 + 6x + 9 = x^2 + 8x + 7$$

$$2x = 2$$

$$x = 1$$

$$(c) x^2 - (k+4)x + 1 = 0$$

for no real roots,

$$\Delta < 0$$

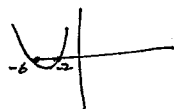
$$\text{i.e. } (k+4)^2 - 4(1)(1) < 0$$

$$k^2 + 8k + 16 - 4 < 0$$

$$k^2 + 8k + 12 < 0$$

$$(k+6)(k+2) < 0$$

$$-6 < k < 2$$



$$(d) V = \pi \int y^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \sec^2 2x dx$$

$$= \pi \left[\tan 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\tan \frac{\pi}{2} - \tan \frac{\pi}{4} \right]$$

$$= \frac{\pi}{2} [\sqrt{3} - 1] u^3$$

Question 7

$$(a) (i) \text{ Focal length} = 2$$

$$\therefore \text{ vertex } (5, 1)$$

$$(ii) (x-5)^2 = -4(2)(y-1)$$

$$(x-5)^2 = -8(y-1)$$

$$(iii) \frac{1}{d} = \left| \frac{2(5) - 1(-1) - 1}{\sqrt{2^2 + (-1)^2}} \right|$$

$$= \frac{10}{\sqrt{5}}$$

$$= \frac{10\sqrt{5}}{5}$$

$$= 2\sqrt{5} \text{ units}$$

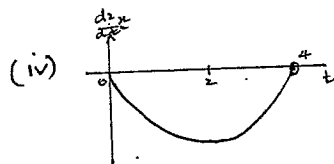
$$(b) (i) t=2, t=5$$

$$(ii) t=2$$

$$(iii) d = \text{area under curve}$$

$$= 2 \times \left(2\frac{2}{3} + 1\right)$$

$$= 7\frac{1}{3}$$

Question 8

$$(a) y = e^{3x}$$

$$\frac{dy}{dx} = 3e^{3x} \text{ at } x=0$$

$$= 3e^0$$

$$= 3$$

$$\text{When } x=0$$

$$y = e^{3 \times 0}$$

$$= 1$$

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$$y-1 = 3(x-0)$$

$$y = 3x + 1$$

$$(b) (i) \frac{d}{dx}(x \ln x - x)$$

$$= x \cdot \frac{1}{x} + 1 \cdot \ln x - 1$$

$$= 1 + \ln x - 1$$

$$= \ln x$$

$$(ii) A = \text{rectangle} - \int \ln x dx$$

$$= e - 1 - [x \ln x - x]$$

$$= e - 1 - [e \ln e - e - (1 \ln 1 - 1)]$$

$$= e - 1 - [e - e - 0 + 1]$$

$$= e - 2$$

$$(c) \frac{dv}{dt} = -k$$

$$(i) v = -kt + c$$

$$\text{when } t=0$$

$$v = \pi(15)^2 \cdot 30$$

$$= 6750\pi$$

$$\therefore c = 6750\pi$$

$$\therefore v = -kt + 6750\pi$$

$$\text{When } t=4 \quad v = 5250\pi$$

$$5250\pi = -4k + 6750\pi$$

$$4k = 1500\pi$$

$$k = 375\pi$$

$$\therefore v = -375\pi t + 6750\pi$$

$$= \pi(6750 - 375t)$$

$$(b) V = 3375\pi$$

$$3375\pi = \pi(6750 - 375t)$$

$$-375t = -3375$$

$$t = 9 \text{ hours.}$$

Question 9

$$(i) P = 2r + r\theta$$

$$60 = 2r + r\theta$$

$$r(\theta+2) = 60$$

$$r = \frac{60}{(\theta+2)}$$

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \left(\frac{60}{\theta+2} \right)^2 \theta$$

$$= \frac{1800\theta}{(\theta+2)^2}$$



$$(ii) \frac{dA}{d\theta} = \frac{(\theta+2)^2 \cdot 1800 - 1800\theta \cdot 2(\theta+2)}{(\theta+2)^4}$$

$$= \frac{(\theta+2)1800[\theta+2-2\theta]}{(\theta+2)^4}$$

$$= \frac{1800(2-\theta)}{(\theta+2)^3}$$

$$2. \text{ Maximum when } \frac{dA}{d\theta} = 0$$

$$\text{i.e. } \frac{1800(2-\theta)}{(\theta+2)^3} = 0$$

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$$\theta = 2$$

Test:

θ	1	2	3
$\frac{dA}{d\theta}$	+	0	-

$\therefore$ max at $\theta = 2$

$$r = \frac{60}{4} = 15$$

$$A = \frac{1}{2} r^2 \theta = \frac{1}{2} 15^2 \cdot 2 = 225 \text{ mm}^2$$

$$(b) \quad x^2 = 4y \quad \dots (1)$$

$$4x - y - 16 = 0 \quad \dots (2)$$

from (1)

$$y = \frac{x^2}{4}$$

$$\therefore 4x - \left(\frac{x^2}{4}\right) - 16 = 0$$

$$x^2 - 16x + 64 = 0$$

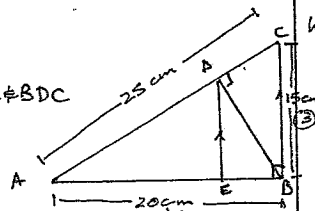
for tangent, $\Delta = 0$ (one)

$$\Delta = 16^2 - 4 \cdot 1 \cdot 64 = 0$$

$$\therefore 4x - y - 16 = 0 \text{ is tangent to}$$

$$x^2 = 4y$$

(c) (i) In $\triangle ABC \cong \triangle BDC$



$$\angle ABC = \angle BDC = 90^\circ \text{ (given)}$$

$$\angle ACB = \angle DCB \text{ (common)}$$

$$\therefore \triangle ABC \cong \triangle BDC \text{ (equiangular)}$$

so sides in ratio

$$\frac{15}{25} = \frac{BD}{20}$$

$$\therefore BD = \frac{15 \times 20}{25}$$

$$= 12 \text{ cm}$$

(ii) $CD = \frac{3}{5} \times 15 = 9 \text{ cm}$

$$\triangle BDE \cong \triangle BCD \text{ (equiangular)}$$

Because $\angle DEB = \angle CDB = 90^\circ$ (-)

$(CB \parallel DE, \text{co-vert } \angle \text{ s equal})$

$\angle BDE = \angle CBD$ (alternate angles are equal; $CB \parallel DE$)

$$\frac{DE}{12} = \frac{12}{15}$$

$$DE = \frac{144}{15} = 9.6 \text{ cm}$$

OR

(1) In $\triangle DCB$

$$x^2 + y^2 = 225 \quad \dots (1)$$

In $\triangle ADE$

$$x^2 + (25 - y)^2 = 400 \quad \dots (2)$$

from (1) $x^2 = 225 - y^2 \quad \dots (3)$

$$(3) \text{ in } (2) \quad 225 - y^2 + 625 - 50y + y^2 = 400$$

$$850 - 50y = 400$$

$$-50y = -450$$

$$y = 9$$

$$x^2 = 225 - 81 = 144$$

$$x = 12$$

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$$\therefore BD = 12 \text{ cm}$$

(ii)

In $\triangle ADE$

$$a^2 + b^2 = 256 \quad \dots (1)$$

In $\triangle DEB$

$$a^2 + (20 - b)^2 = 144 \quad \dots (2)$$

from (1)

$$a^2 = 256 - b^2 \quad \dots (3)$$

(3) into (2)

$$256 - b^2 + 400 - 40b + b^2 = 144$$

$$656 - 40b = 144$$

$$40b = 512$$

$$b = 12.8$$

$$a^2 = 256 - 12.8^2$$

$$= 92.16$$

$$a = 9.6$$

$$\therefore DE = 9.6 \text{ cm}$$

Question 10

$$a_x = -2 \sin t_1, \quad t = 0, \quad v = 2, \quad x = 2$$

(i) $a_x = -2 \sin t_1$

$$v_x = 2 \cos t_1 + c$$

when $t = 0 \quad v = 2$

$$2 = 2 \cos 0 + c$$

$$2 = 2 + c$$

$$\therefore c = 0$$

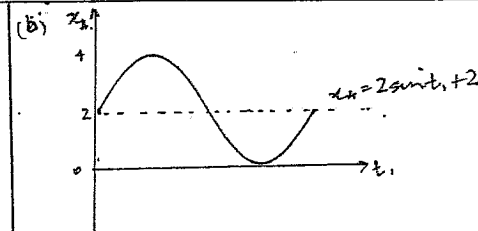
so $v_x = 2 \cos t_1$

$$x_x = 2 \sin t_1 + c$$

$$2 = 2 \sin 0 + c$$

$$c = 2$$

$$\therefore x_x = 2 \sin t_1 + 2$$



(c) $a_x = \cos t_1, \quad t_2 = t_1 - \frac{\pi}{2}$

$$a_x = \cos(t_1 - \frac{\pi}{2})$$

$$= \cos(\frac{\pi}{2} - t_1)$$

$$= \sin t_1$$

$$t = \frac{\pi}{2}, \quad v = 0, \quad x = 1$$

$$v_x = -\cos t_1 + c$$

$$0 = -\cos \frac{\pi}{2} + c$$

$$0 = -1 + c$$

$$c = 1$$

$$\therefore v_x = -\cos t_1$$

$$x_x = -\sin t_1 + c$$

$$1 = -\sin \frac{\pi}{2} + c$$

$$1 = -1 + c$$

$$c = 2$$

$$\therefore x_x = 2 - \sin t_1$$

(d) when $t_1 = \frac{\pi}{2}$

$$x_x = 2 \sin \frac{\pi}{2} + 2$$

$$= 4$$

$$x_x = 2 - \sin \frac{\pi}{2}$$

$$= 2 - 1$$

$$= 1$$

$$\therefore \text{Distance b/w} = 3 \text{ cm}$$

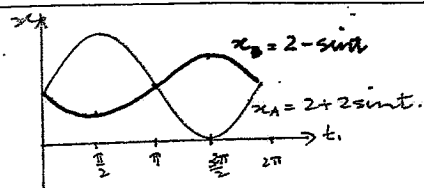
(e) $2 + 2 \sin t_1 = 2 - \sin t_1$

$$3 \sin t_1 = 0$$

$$\sin t_1 = 0$$

$$t_1 = 0, \pi, 2\pi$$

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from $t = \pi \rightarrow t = 2\pi$

$$d_A = 4\text{cm}$$

$$d_B = 2\text{cm}$$

$\therefore$ Difference = 2cm.

Alternate (c)

$$a_B = \cos t_2$$

$$\therefore v_B = \sin t_2 + C$$

when $t_2 = 0$

$$v_B = 0$$

$$0 = \sin 0 + C$$

$$C = 0$$

$$\therefore v_B = \sin t_2$$

$$t_2 = t_1 - \frac{\pi}{2}$$

$$= \sin\left(t_1 - \frac{\pi}{2}\right)$$

$$= \sin\left(-\left(\frac{\pi}{2} - t_1\right)\right)$$

$$= -\sin\left(\frac{\pi}{2} - t_1\right)$$

$$v_B = -\cos t_1$$